

HMS Strategy & Operations

COMPREHENSIVE SYSTEM ARCHITECTURE & ROLE-BASED ACCESS REPORT

1. System Module Architecture

A breakdown of the core operational modules required for a modern clinical environment.

Command Center

Real-time analytics and hospital-wide KPI tracking.

User Management

Secure RBAC and personnel credentialing.

Patient Directory

Centralized registry for longitudinal medical records.

EMR / EHR

Electronic health records and clinical documentation.

Revenue Cycle (Billing)

Invoicing, insurance claims, and financial auditing.

Pharmacy Management

Inventory, dispensing, and prescription integration.

Diagnostics & Lab

Order management and digital report delivery.

Facility Management

Bed allocation, ward occupancy, and cleaning logs.

Proposed Expansion Modules

Telemedicine Hub

Virtual consultations and remote patient monitoring.

Insurance & TPA

Automated claim processing and cashless approvals.

Ambulance Dispatch

GPS tracking for emergency response fleets.

Asset Maintenance

Clinical equipment uptime and service tracking.

2. Detailed Role-Based Access Control (RBAC)

The following table outlines the module-level permissions for each clinical and administrative role.

ROLE	PERMITTED MODULES	CORE RESPONSIBILITY
Admin	All Modules (Command Center, User Management, Reports, etc.)	Full System Administration
Doctor	EMR, Appointments, Calendar, Lab, Blood Bank, Telemedicine	Clinical Diagnosis & Treatment
Nurse	EMR (Vitals), Patient Registry, Beds, Blood Bank, Pharmacy (View)	Nursing Care & Ward Mgmt
Reception	Patient Registry, Appointments, Billing, Beds, Ambulance Dispatch	Patient Onboarding & Logistics
Pharmacist	Pharmacy, Inventory, Medicine Stock, Expiry Alerts	Medication Safety & Logistics
Lab Technician	Diagnostics & Lab, Patient Directory (View), Equipment Maintenance	Diagnostic Testing & Reporting
Accountant	Revenue Cycle (Billing), Insurance/TPA, Financial Reports	Financial Integrity & Claims

3. The Patient Journey Workflow

1

Registration & Triage

Role: Receptionist & Nurse

Profile creation followed by recording of vitals and priority assessment.

2

Clinical Consultation

Role: Doctor

Examination, diagnosis, and generation of orders (Labs/Meds).

3

Diagnostics & Investigation

Role: Lab Technician

Sample collection and digital result synchronization to EMR.

4

Financial Settlement

Role: Accountant / Billing Clerk

Invoice generation, TPA claim processing, and payment collection.

5

Medication & Exit

Role: Pharmacist

Prescription fulfillment and dispensing with dosage guidance.